

ATTACHMENT ()
TECHNICAL ORDERING DATA (TOD)

CONTRACT No: _____
ITEM: _____

SECTION B - SUPPLIES/SERVICES:

NATIONAL STOCK NUMBER (NSN): 1H 4710-00-455-0197 X2

ITEM NAME: PIPE, 1 IN, SMLS, NI-CU

PART NUMBER: MIL-DTL-23520,1.315OD

QUANTITY: 19,656 INCHES

SPECIAL CHARACTERISTICS: NPS 1 INCH, SCH 80, Seamless, LEVEL 1. (1.315" OD, 0.179" nominal wall thickness)

SECTION C - DESCRIPTION/SPECIFICATIONS:

C.1. Manufacture per the attached Individual Repair Part Ordering Data (IRPOD), as modified by this contract.

C.1.1. The order of precedence for any conflicting documentation/requirements shall be as follows:

- (a) The contract (including attachments, modifications, amendments, etc.).
- (b) The IRPOD.
- (c) Drawings (if any) referenced by the IRPOD.
- (d) Specifications referenced by the IRPOD or drawing.

C.1.2. Manufacture in accordance with MIL-DTL-23520, except as modified herein. Ordering data for paragraph 6.2 of MIL-DTL-23520 is as follows:

- (a) "Tube and Pipe, Nickel Copper Alloy, Seamless, Air Melted (UNS N04400)", MIL-DTL-23520E (SH) of 30 August 2021.
- (b) Class IV (4), Finish D (Descaled), Size Schedule and Quantity to be as in Section B of the schedule.
- (c) Not Applicable
- (d) For 1/2" thru 2" NPS Sizes: furnish to outside diameter and wall thickness.
- (e) Class IV Lengths per Paragraph 3.3.2.3(b), except maximum lengths of 20 feet in lieu of 24 feet are preferable.
- (f) Material making requirements of Paragraph 3.6.2 apply along with the requirements of C.5 below.
- (g) Vendor to perform metal removal when applicable in lieu of purchaser
- (h) Liquid Penetrant Required

C.1.3. Ultrasonically inspect pipe per MIL-DTL-23520 and MIL-STD-2132.

- (a) The vendor shall notify the purchasing agent four (4) weeks in advance of ultrasonic testing to allow participation of Portsmouth Naval Shipyard personnel in witnessing the ultrasonic tests.

C.1.4. MIL-DTL-23520 paragraph 4.4.2.1 Clarification: Each finished pipe and tube, regardless of size, shall be ultrasonically examined over 100% of the Outside Diameter surface to assure compliance with the specified Wall Thickness requirements.

C.2. APPLICABLE DOCUMENTS:

C.2.1. Section 2.0 of the IRPOD lists applicable documents to be used (e.g., drawings, specifications, standards, etc.) and their required version level (e.g., revisions, amendments, change notices, interim changes, etc.). The following list of documents and version information takes precedence over version information (including the lack of any version information) identified by the IRPOD:

- (a) MIL-DTL-23520E, TUBE AND PIPE, NICKEL-COPPER ALLOY, SEAMLESS AIR MELTED (UNS N04400)

- (b) MIL-STD-792F CHANGE 1; IDENTIFICATION MARKING REQUIREMENTS FOR SPECIAL PURPOSE COMPONENTS
- (c) MIL-STD-129R; MARKING FOR SHIPMENT AND STORAGE
- (d) ASTM D3951-18; STANDARD PRACTICE FOR COMMERCIAL PACKAGING
- (e) MIL-STD-2132F; NONDESTRUCTIVE EXAMINATION REQUIREMENTS FOR SPECIAL APPLICATIONS

C.2.2. If no document version is identified (anywhere in the contract or referenced documents), then use the latest version identified in the Acquisition Streamlining and Standardization Information System (ASSIST) at the time of contract award; ASSIST information is available at the following website:

<https://quicksearch.dla.mil>. If no document version information is available in ASSIST, then any version of the document may be used provided its version is dated 1 November 1969 or later.

C.2.3. In any case, the vendor shall use only one version in its entirety (i.e., the vendor shall not use portions of different versions of a document).

C.3. DATA REQUIREMENTS:

C.3.1. Material Certification; See Exhibit A, Data Item No. A001.

C.3.2. Procedure Approvals; See Exhibit A, Data Item No.'s A002, A003, and A004. The vendor shall provide, and obtain approval for, the procedures identified below prior to their use. If the vendor considers a required submittal inapplicable to this procurement, the vendor shall submit a statement justifying the exclusion of the required submittal.

- (a) ULTRASONIC TEST PROCEDURE - Procedure (per MIL-STD-2132 – for both soundness and thickness), including acceptance criteria (per MIL-DTL-23520), shall be submitted. Ultrasonic calibration standard certification shall also be submitted for review and approval (this may be submitted no later than two weeks prior to scheduling the ultrasonic inspection witness).
- (b) ALLOY IDENTITY – Each tube and pipe shall be alloy identity tested for the correct generic metallurgical group as defined in MIL-STD-2132. The test shall be performed after the final marking required by 3.6, unless the material passes directly from the alloy identity device to the marking device.
- (c) LIQUID PENETRANT TEST PROCEDURE – Procedure conforming to the applicable requirements of MIL-STD-2132 shall be used for liquid penetrant test examination.

C.4. CLEANLINESS REQUIREMENTS:

C.4.1. Cleanliness requirements shall be per Section 3.2 of the IRPOD and Paragraph 5.1 of MIL-STD-23520.

C.5. MATERIAL MARKING (LEVEL 1 MATERIAL):

C.5.1. Each piece shall be permanently marked on each end with the information of (a), (b), (c), and (d) below using the permanent marking methods of MIL-STD-792 or the permanent marking method specified in the material specification. If any marking specified below cannot be applied due to space limitation, a waiver request shall be submitted to the contracting officer.

- (a) The Material Designator/Code "NCF".
- (b) The manufacturer's name, trademark, or symbol.

- (c) The vendor traceability number traceable to all manufacture test and inspection information and to a unique set of physical, mechanical, and chemical test results which represent the material used to make the piece.
- (d) Any information required by the specification or drawing (if applicable) in addition to that required by (a), (b), and (c) above shall be marked either per the specification/drawing or on each piece at each end per the permanent marking methods authorized above. Any Material Designator/Code listed in the specification or drawing (if applicable) need not be marked on material, if it is different from that specified by (a) above.

C.5.2. The location of the markings shall be on the outer diameter on the end(s) of each piece. Die stamp markings shall not be used for permanently marking pipe or tube per MIL-STD-792.

C.5.3. If feasible, consolidate the placement of the permanent marking(s) to accommodate the marking of a nine digit final acceptance marking at Portsmouth Naval Shipyard. Do not compromise the legibility of the permanent markings to meet this request.

C.6. NAVSEA HQ C-2-0023 Mercury Control.

SECTION D - PRESERVATION/PACKAGING/PACKING:

D.1. Marking requirements for packaging shall be per Section 5.0 of the IRPOD.

D.2. Preservation, Packaging, and Packing shall be in accordance with ASTM D3951 and MIL-DTL-23520 paragraph 5.2, as modified below:

- (a) The material shall be shipped in wooden boxes. For 6-inch OD and larger pipe, the wooden boxes shall be constructed of 2-inch thick by 6-inch wide minimum planks nailed or screwed together. The exterior of the boxes shall be banded with metal straps.

D.3. The use of MIL-STD-2073-1 packaging codes are not required for this contract.

SECTION E - INSPECTION ACCEPTANCE:

E.1. Government inspection is required during manufacture and prior to shipment from this facility. Upon receipt of this order, promptly notify the Representative of the Government office who normally serves your facility so that appropriate planning for Government inspection can be accomplished. In the event a Representative of the Government office cannot be located, the purchasing agent should be notified promptly. Provide the location where Procurement Quality Assurance (PQA) of the supplies will be made along with the organization that will be performing PQA.

E.2. All non-conformances to contract requirements affecting product and/or documentation shall require procuring activity waiver approval to accept.

E.3. FAR 52.246-11, Higher-Level Contract Quality Requirements

Supplemented by:

NAVSEA HQ H-246-H001 Calibration System Requirements (NAVSEA)

NAVSEA HQ E-246-H020 Quality Management System Requirements (NAVSEA)

ATTACHMENT ()
TECHNICAL ORDERING DATA (TOD)

CONTRACT No: _____
ITEM: _____

NAVSEA HQ C-246-H002 Government use of Contractor's Inspection Equipment (NAVSEA)
(MIL-I-45208 or ISO 9001/9002/9003 with STR-9000 Supplement A or B or C)

E.4. FAR 52.246-2 Inspection of Supplies-Fixed-Price (Aug 1996).

GOVERNMENT INSPECTION CHECK-OFF LIST (GICOL)

NUCLEAR MATERIAL ORDERING – TECHNICAL CODE ACTION PART - A			PROCUREMENT QUALITY CONTROL ACTION PART - B	
To: <i>Code 2305.1</i>		Engineering Concurrence and Date <i>N/A</i>	Contract Number and Date	
Document Number <i>26RX19764</i>	QA Code <i>QA-A (LEVEL I)</i>		Code 2305.1 ADDED ATTRIBUTE	REMARKS
Nomenclature <i>NSN: 1H 4710-00-455-0197 X2</i>				
ATTRBT	REQD	IDENTIFICATION BRIEF		
01	☒	DOCUMENTATION		
02	☒	DAMAGE		
03	☒	PRESERVATION PACKAGING, PACKING AND LABELS		
04	☒	VISUAL		
05	☒	MATERIAL IDENTIFICATION		
06	☒	MATERIAL VERIFICATION TESTS		
07	☐	RADIOGRAPHY		
08	☐	MAGNETIC PARTICLE TEST		
09	☒	ULTRASONIC/EDDY CURRENT TEST		ULTRASONIC
10	☒	LIQUID PENETRANT TEST		
11	☐	OPERATIONAL OR FUNCTIONAL TEST		
12	☒	PRESSURE TEST		
13	☐	ELECTRICAL ELECTRONIC TEST		
14	☐	MISSING, WRONG, IMPROPERLY ASSEMBLED PARTS		
15	☒	DIMENSIONS		
16	☐	WELDING		
19	☐	FINISH		
20	☐	SHELF - LIFE		
22	☐	MERCURY FREE		
23	☐	SPECIAL PROCESS PROCEDURE APPROVAL		
24	☒	MANUFACTURING PROCESS		SEAMLESS
25	☐	DESIGN EVALUATION TESTS		
61	☐	QUANTITATIVE CHEMICAL ANALYSIS		
62	☐	SEMI-QUANTITATIVE CHEMICAL ANALYSIS		
63	☒	GENERIC MATERIAL VERIFICATION		
64	☐	MECHANICAL PROPERTIES VERIFICATION		
65	☐	MECHANICAL PROPERTIES INDICATOR (HARDNESS TEST)		
Procurement Quality Assurance Invoked ☒ Yes ☐ No			Procurement Quality Assurance Invoked ☐ Yes ☐ No	
Code 2305.3/.4 Signature and Date 		Code 2305.4	Preparer Signature and Date	Code

NOTES:

(1) SEE ALSO "MANDATORY INSPECTIONS" AND ATTACHMENT TO GICOL.

MANDATORY INSPECTIONS

This attachment provides additional GICOL guidance and sampling for attributes that have been added to the DCMA source inspections for this contract by Code 2305.4.

Where IAC 06 is invoked, ensure that the instructions to the QAR contain a requirement to verify that procedures requiring approval have been stated to be used to perform the applicable process or testing.

Additional NDT attributes which are added to the GICOL by Code 2305.4 in addition to the requirements of the IRPOD MI shall be sampled as follows (IACs 07-13, 15 and 63):

“Witness the first production test, and testing performed on 20% of the remainder of the lot. Verify applicable procedures have been approved if required by the contract”

Additional Non-NDT attributes (IACs 14, 16, 20, 22, 23-25, 61, 62, 64, and 65):

“Record review 100% of the objective quality evidence (e.g. inspection/test reports, procedure approvals etc.) to verify conformance with applicable contract/specification requirements.”

To assist the QAR in review of material markings (IAC 05), the following provides a list of the minimum expected permanent markings to be present on the material:

- (a) The Material Designator/Code “NCF” *[STR-505, Para. 5.2, and Contract section C.5.1(a)]*
- (b) The Manufacturer’s name, trademark, or symbol *[STR-505, Para. 5.2]*
- (c) The Vendor's Traceability Number (Note: Needs to match that provided on the Vendor's certifications) *[STR-505, Para. 5.2]*
- (d) The specification number and revision letter *[MIL-DTL-23520E Para. 3.6.3(d)]*
- (e) Class, Size and Schedule or Wall Thickness, as applicable *[MIL-P-23520E, Para. 3.6.3(d)]*

To assist the QAR in certification review (IAC 06), the following provides a list of the minimum expected certifications to be provided with the material:

- (a) Certification of Compliance
- (b) Chemical Analysis
- (c) Mechanical Properties
- (d) Seamless
- (e) MIL-DTL-23520 Test Report Certification, to include:
 - i. Ultrasonic Testing (wall thickness and flaw)
 - ii. Bending Test.
 - iii. Visual.
 - iv. Dimensional.
 - v. Hydrostatic Test
 - vi. Alloy Identity
 - vii. Liquid Penetrant Test.